

HW3 Numerical Methods

- 1) What function, $y = mx + b$ approx $P(x) = \sin(x)$ best close to $x = \pi/4$. How does this relate to Newton's Method?
- Linear Approx given by $P_2(x)$ Taylor Polynomial about $x = \pi/4$
 $P_2(x) = f(x) + f'(x)(x - x_0)$

$$f(\pi/4) = \sin(\pi/4) = \frac{\sqrt{2}}{2} \quad P_2(x) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}(x - \frac{\pi}{4})$$

$$f'(\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2} \quad \text{get } y = mx + b$$

$$y = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}x - \frac{\pi\sqrt{2}}{8}$$

$$= \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}(1 - \frac{\pi}{4}) \quad a = \frac{\sqrt{2}}{2} \quad b = \frac{\sqrt{2}}{2}(1 - \frac{\pi}{4})$$

- $y = \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}(1 - \frac{\pi}{4})$ is a tangent line @ $\sin(\pi/4)$.
- Newton's Method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ is used to find roots, uses tangent lines iteratively.
- The problem approximates a point whereas Newton's Method approx. a root, but both use tangent lines

2) Locate Root of $f(x) = e^{-x} - \cos(x)$ near $\frac{\pi}{4}$
 $x_0 = \frac{\pi}{4}$ (Root To Find)

$$f(\pi/4) = e^{-(\pi/4)} - \cos(\pi/4) \approx -0.2529 \quad \text{Init conditions}$$

$$f'(x) = -e^{-x} - \sin(x) = -e^{-x} - \sin(x)$$

$$f'(\pi/4) = -e^{-(\pi/4)} - \sin(\pi/4) \approx -0.6529$$

$$x_1 = \frac{\pi}{4} - \frac{-0.2529}{-0.6529} = \frac{\pi}{4} + 1 \approx 1.7854$$

$$x_2 = 1.7854 - \frac{f(1.7854)}{f'(1.7854)} \approx 1.3132$$

$$x_3 = 1.3132 - \frac{f(1.3132)}{f'(1.3132)} \approx 1.2915$$

$x = 1.2915$ is close to Denote Root of 1.2927, More iterations will be more accurate

HW#3 Numerical Methods

- 3) For what starting values (x_0) will Newton's Method converge if $f(x) = 0$, $f(x) = \frac{x^2}{1+x^2}$

Find Root

$$0 = \frac{x^2}{1+x^2} \Rightarrow x^2 = 0$$

root at $x=0$

Intervals of increase/decrease

$$f'(x) = \frac{(1+x^2)(2x) - x^2(2x)}{(1+x^2)^2}$$

$$= \frac{2x}{(1+x^2)^2}$$

implies decreasing on $(-\infty, 0)$

increasing on $(0, \infty)$

$$f''(x) = \frac{2(-3x^2+1)}{(1+x^2)^3}$$

$$f''(x) = 0 \text{ @ } x = \pm \frac{1}{\sqrt{3}}$$

It can also be noted that the further away from 0 you go, the worse the rate of convergence since the functions range is $[0, 1]$, implying flatter and flatter T-lines

By Observation: Function is always positive and has one root. Function is also Even.

Newton's Method will converge for all non-zero values.

→ This is because the function is even with only one root, and the increase and decrease imply that there will be no "unhappy" case (See Kincaid's pg 131).

→ Intervals of Increase/decrease also imply no "Flat Spot" case. (Kincaid 131) and No "cycle" case.